A Mechanistic Model for Radiation-Induced Segregation Controlled Grain-Boundary Oxidation in Austenitic Stainless Steels

Zhuo Wang Md Shahrier Hasan Jaime Marian

October 16, 2026

1 Introduction

Irradiation-assisted stress corrosion cracking (IASCC) is a major degradation mode of austenitic stainless steel core internals in light-water reactors [1]. In irradiated stainless steels exposed to high-temperature water, intergranularoxidation weakens grain boundaries and precedes intergranular crack initiation [2], and grain-boundary oxidation has been identified as a necessary precursor of IASCC at low doses [3]. In 304 stainless steel (304SS) exposed to pressurized water reactor (PWR) primary water, the extent of this oxidation depends on the chemical changes introduced by prior irradiation. Deng et al. [4] found that increasing the proton irradiation dose produced deeper intergranular oxidation in 304 nuclear-grade stainless steel after water exposure, accompanied by a decrease in the Cr content of the intergranular oxide. Cr depletion is not the only chemical change that matters. Using model alloys with Si and Cr contents chosen to mimic the segregated boundary, Du et al. [5] showed that Si enrichment promotes intergranular oxidation and cracking even when Cr is not depleted: Si forms SiO_2 that is unstable in high-temperature water and dissolves preferentially, leaving a porous oxide that provides pathways for oxygen diffusion along grain boundaries. These observations connect irradiation to both the chemistry and the penetration depth of the intergranular oxide. Turning them into a predictive relationship requires a description of how the changing oxide composition and structure affect oxygen transport and, consequently, the advance of the oxidation front.

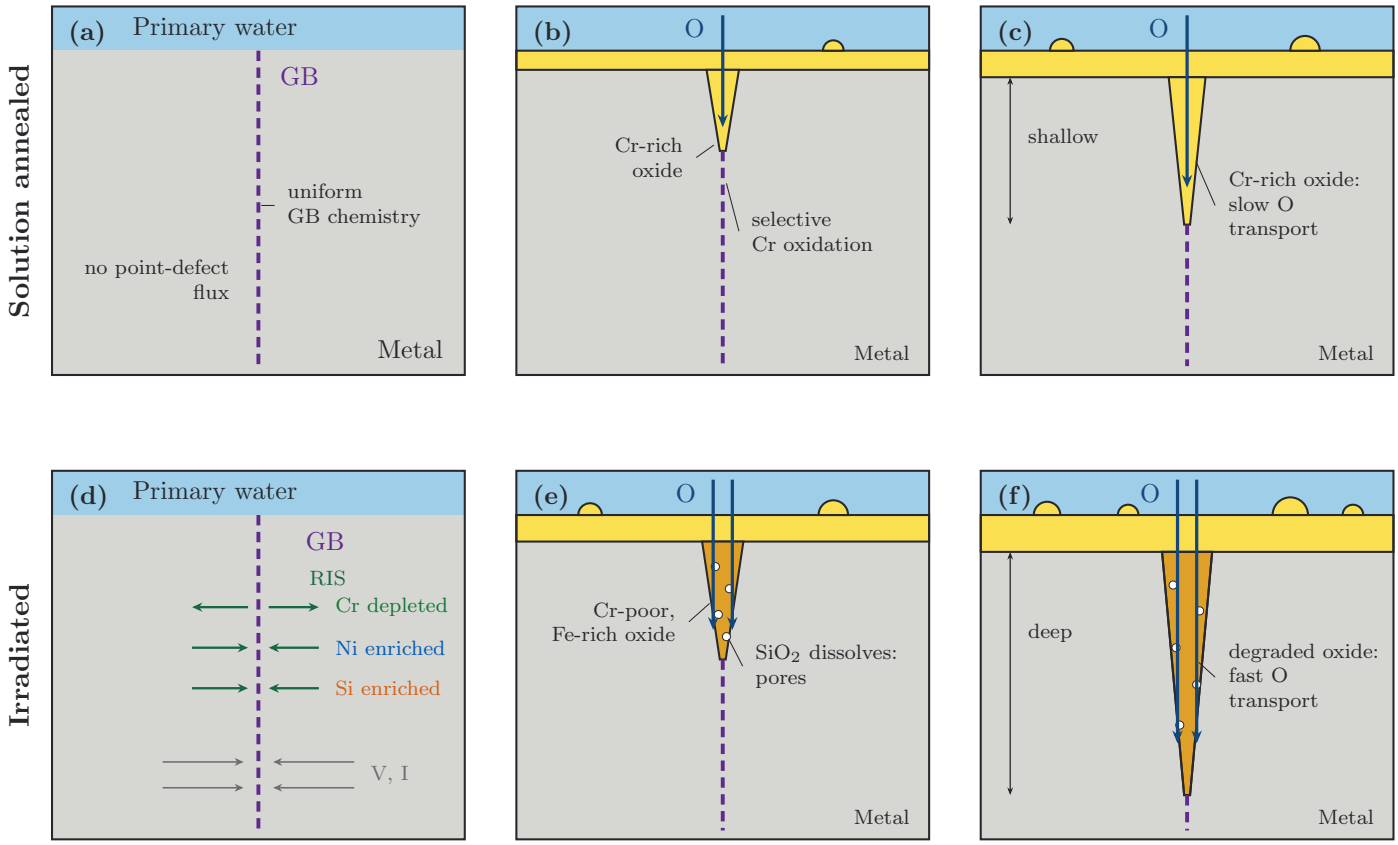


Figure 1: Physical picture of intergranular oxidation in primary water without (a–c) and with (d–f) prior irradiation. (a) Solution-annealed boundary with uniform chemistry. (b) Oxygen entering at the boundary mouth oxidizes Cr selectively and forms a Cr-rich intergranular oxide. (c) The Cr-rich oxide limits oxygen transport along the boundary and the penetration stays shallow. (d) Irradiation drives point defects to the boundary and the resulting radiation-induced segregation depletes Cr and enriches Ni and Si. (e) On exposure the Cr-depleted boundary forms a Cr-poor, Fe-rich oxide, and the segregated Si oxidizes and dissolves, leaving pores. (f) The degraded oxide transports oxygen faster along the boundary and the oxide penetrates deeper.

Radiation-induced segregation (RIS) establishes the local alloy composition from which the intergranular oxide develops. Coupling between solute and irradiation-generated point-defect fluxes redistributes alloying elements near grain boundaries. Measurements on irradiated 304 nuclear-grade stainless steel reveal Cr depletion and Ni and Si enrichment at the grain boundary [6]. Subsequent oxidation selectively consumes alloying elements, changing both the oxide composition and the adjacent metal, where Ni enrichment has also been observed [6]. The lower Cr content of the oxide formed on irradiated boundaries has been associated with reduced resistance to oxygen diffusion [4].

Si oxidation and dissolution alter this transport resistance by changing the structure through which oxygen moves. The role of Si identified by Du et al. [5] established with model alloys of systematically varied Cr and Si contents in hydrogenated water, which allowed the contributions of Cr depletion and Si enrichment to be separated. They attributed the greater oxide degradation and cracking susceptibility at higher Si content to dissolution of SiO_2 in the inner oxide and the resulting porous, defective film, and their STEM analysis of intergranular oxides indicated faster grain-boundary diffusion at higher Si content. Related microscopy on proton-irradiated 316L identified Si enrichment at oxide tips and porosity behind them, consistent with preferential Si oxidation followed by dissolution [3]. Although 316L differs from 304SS in bulk composition, this observation is used here as mechanistic support rather than as a quantitative reference; the same oxidation and dissolution of segregated Si appears as a step in the IASCC mechanism summarized by Was et al. [2]. Together, these studies provide a basis for including dissolution-generated free volume in the transport description. With the dependence on Cr content, they motivate representing the evolving oxide composition and free volume through a single effective oxygen diffusivity, which connects the segregation of multiple elements to a common quantity governing oxygen transport.

Each half of this sequence has been modeled separately, and the two halves are not equally developed. The segregation half is settled enough that the boundary composition can be computed rather than only measured. Since the inverse-Kirkendall and rate-theory descriptions of RIS in concentrated alloys [**marwick1978segregation**, **wiedersich1979theory**], the composition profile at a grain boundary has been obtained from the coupled fluxes of point defects and alloying elements, with Fe–Cr–Ni parameterizations that reproduce the measured dose and temperature dependence of Cr depletion

and Ni enrichment [7]. Current models go further and resolve the sink itself, predicting segregation at grain boundaries and dislocation cell walls [8] and the conditions under which Cr switches from enrichment to depletion [9]. What all of these models return is the segregation profile. None carries that profile into what the boundary does next: the profile is an end point, not the initial condition for a subsequent process.

On the oxidation side, the dependence of intergranular penetration depth on dose and exposure time has been characterized experimentally [4, 10], but existing kinetic models of grain-boundary oxidation do not respond to the boundary chemistry. Models that resolve oxygen transport along the boundary treat the oxygen diffusivity as a single constant: in analytical or semi-analytical solutions that give the penetration depth in closed form [11–13], in finite-element descriptions in which it is fitted to measured oxide depths through a parabolic law [14, 15] or prescribed together with a constant reaction rate [16], and in phase-field simulations of a single stoichiometric chromia intrusion [17, 18]. In none of these does the diffusivity depend on the composition of the oxide that has formed. Where the oxide identity does control transport, in multilayer descriptions of planar scales [19, 20] and in the mechanistic treatment of uniform corrosion in high-temperature water [21], the controlling phase is fixed at the outset rather than selected by the local alloy composition. The boundary composition enters only after conversion into an empirical parameter, as a Cr-content-dependent dissolution rate in sensitized steels [22] or as a sensitization index that increases with neutron fluence in the slip-oxidation prediction of crack growth [23]; crack-growth models that require a grain-boundary oxidation law borrow rate constants measured on bulk surfaces [24]. Thermodynamic selection of the oxide phases from the local alloy composition has been implemented for a Ni-base superalloy [25], but with a constant oxygen diffusivity, the bulk composition at the boundary, and without the Cr oxide that limits transport. No model takes the irradiation-modified boundary composition as input and lets the resulting oxide composition set the oxygen transport that governs further penetration.

To our knowledge, no model connects the segregated composition to the advance of the oxidation front through a transport quantity computed from both. Establishing this connection requires tracking how the segregated alloy supplies the interfacial reactions, how those reactions change the oxide, and how the resulting diffusivity affects subsequent oxygen delivery. Segregation profiles, oxide composition, interfacial Ni enrichment, and penetration depth

then provide complementary observables for assessing the coupling.

Here, we develop a coupled RIS and grain-boundary oxidation model for 304SS, represented by the Fe–Cr–Ni–Si system. The model links defect-mediated solute redistribution and selective interfacial consumption to an effective oxygen diffusivity, D_{eff}, determined by the evolving oxide composition and free volume. Within this description, increasing the Cr content of the oxide lowers the effective diffusivity, whereas Si oxidation followed by complete dissolution contributes free volume that enhances transport. The same D_{eff} enters both oxygen transport along the grain boundary and oxygen delivery across the oxide to the metal interface. The model therefore connects RIS to oxidation kinetics through the resulting change in oxygen transport, with metal consumption feeding back on the local composition. The complete-dissolution treatment is a simplification motivated by the mechanism discussed by Du and co-workers [5]. We examine the dose and exposure-time dependence of oxide penetration under sequential irradiation and water exposure, using separate calibration of the RIS and oxidation components and a common set of oxidation parameters across doses. The framework also permits predictions of concurrent RIS and oxidation within the same transport description.

References

[1] Gary S. Was, Yugo Ashida, and Peter L. Andresen. "Irradiation-Assisted Stress Corrosion Cracking". In: *Corrosion Reviews* 29.1–2 (2011), pp. 7–49. ISSN: 2191-0316 0334-6005. DOI: 10.1515/corrrev.2011.020.

[2] G. S. Was et al. "How Irradiation Promotes Intergranular Stress Corrosion Crack Initiation". In: *Progress in Materials Science* 143 (2024). ISSN: 00796425. DOI: 10.1016/j.pmatsci.2024.101255.

[3] Wenjun Kuang et al. "A High-Resolution Characterization of Irradiation-Assisted Stress Corrosion Cracking of Proton-Irradiated 316L Stainless Steel in Simulated Pressurized Water Reactor Primary Water". In: *Corrosion Science* 199 (May 2022), p. 110187. ISSN: 0010-938X. DOI: 10.1016/j.corsci.2022.110187. URL: http://dx.doi.org/10.1016/j.corsci.2022.110187.

[4] Ping Deng et al. “Effect of Irradiation on Corrosion of 304 Nuclear Grade Stainless Steel in Simulated PWR Primary Water”. In: *Corrosion Science* 127 (2017), pp. 91–100. ISSN: 0010938X. DOI: 10.1016/j.corsci.2017.08.010.

[5] Donghai Du et al. “Silicon Segregation Enhances IASCC of Austenitic Stainless Steels via Oxide Degradation: Decoupling from Chromium Depletion Effects”. In: *Corrosion Science* 258 (2026). ISSN: 0010938X. DOI: 10.1016/j.corsci.2025.113424.

[6] P. Deng, Q. Peng, and E. H. Han. “Grain Boundary Oxidation of Proton-Irradiated Nuclear Grade Stainless Steel in Simulated Primary Water of Pressurized Water Reactor”. In: *Sci Rep* 11.1 (Jan. 14, 2021), p. 1371. ISSN: 2045-2322 (Electronic) 2045-2322 (Linking). DOI: 10.1038/s41598-020-80600-x.

[7] T.R. Allen and G.S. Was. “Modeling Radiation-Induced Segregation in Austenitic Fe–Cr–Ni Alloys”. In: *Acta Materialia* 46.10 (June 1998), pp. 3679–3691. ISSN: 1359-6454. DOI: 10.1016/s1359-6454(98)00019-6. URL: http://dx.doi.org/10.1016/S1359-6454(98)00019-6.

[8] Sourabh Bhagwan Kadambi et al. “Phase-Field Modeling of Radiation-Induced Composition Redistribution: An Application to Additively Manufactured Austenitic Fe–Cr–Ni”. In: *Computational Materials Science* 255 (2025). ISSN: 09270256. DOI: 10.1016/j.commatsci.2025.113895.

[9] Russell Oplinger et al. “Radiation-Induced Segregation in Dilute Fe–Cr: A Rate-Theory Framework for the Cr Enrichment–Depletion Transition at the Grain Boundary”. In: *Journal of Applied Physics* 139.15 (2026). ISSN: 0021-8979 1089-7550. DOI: 10.1063/5.0323296.

[10] Ping Deng et al. “Corrosion Behavior and Mechanism of Irradiated 304 Nuclear Grade Stainless Steel in High-Temperature Water”. In: *Acta Metallurgica Sinica (English Letters)* 34.2 (2020), pp. 174–186. ISSN: 1006-7191 2194-1289. DOI: 10.1007/s40195-020-01123-y.

[11] J. C. Fisher. “Calculation of Diffusion Penetration Curves for Surface and Grain Boundary Diffusion”. In: *Journal of Applied Physics* 22.1 (1951), pp. 74–77. ISSN: 0021-8979 1089-7550. DOI: 10.1063/1.1699825.

[12] Zhijie Xu et al. "Modeling Selective Intergranular Oxidation of Binary Alloys". In: *The Journal of Chemical Physics* 142.1 (Jan. 7, 2015), p. 014704. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.4905226. URL: https://pubs.aip.org/jcp/article/142/1/014704/193724/Modeling-selective-intergranular-oxidation-of (visited on 10/07/2026).

[13] H. E. Evans, H. Y. Li, and P. Bowen. "A Mechanism for Stress-Aided Grain Boundary Oxidation Ahead of Cracks". In: *Scripta Materialia* 69.2 (July 1, 2013–), pp. 179–182. ISSN: 1359-6462. DOI: 10.1016/j.scriptamat.2013.03.026.

[14] L. G. Zhao. "Modeling of Oxygen Diffusion Along Grain Boundaries in a Nickel-Based Superalloy". In: *Journal of Engineering Materials and Technology* 133.3 (June 24, 2011). ISSN: 0094-4289. DOI: 10.1115/1.4003777. URL: http://dx.doi.org/10.1115/1.4003777.

[15] A. Karabela et al. "Oxygen Diffusion and Crack Growth for a Nickel-Based Superalloy under Fatigue-Oxidation Conditions". In: *Materials Science and Engineering: A* 567 (Apr. 2013), pp. 46–57. ISSN: 0921-5093. DOI: 10.1016/j.msea.2012.12.088. URL: http://dx.doi.org/10.1016/j.msea.2012.12.088.

[16] Yoshiteru Aoyagi and Yoshiyuki Kaji. "Crystal Plasticity Simulation Considering Oxidation along Grain Boundary and Effect of Grain Size on Stress Corrosion Cracking". In: *MATERIALS TRANSACTIONS* 53.1 (2012), pp. 161–166. ISSN: 1345-9678 1347-5320. DOI: 10.2320/matertrans.MD201126.

[17] Victor de Rancourt. "Modelling the Oxidation of Polycrystalline Austenitic Stainless Steels Using a Phase Field Approach Coupled with Mechanics". Paris: MINES ParisTech, PSL Research University, 2015. URL: https://pastel.hal.science/tel-01253941.

[18] Victor de Rancourt et al. "Viscoplasticity and Phase Field Modelling of the Delayed Cracking of Oxide Intrusions at Grain Boundaries". In: *Journal of the Mechanics and Physics of Solids* 205 (Dec. 2025), p. 106300. ISSN: 0022-5096. DOI: 10.1016/j.jmps.2025.106300. URL: http://dx.doi.org/10.1016/j.jmps.2025.106300.

[19] P. L. Harrison. "The Ratios of Layer Thicknesses in a Multiphase Oxide Scale". In: *Oxidation of Metals* 22.1–2 (Aug. 1984), pp. 35–43. ISSN: 0030-770X, 1573-4889. DOI: 10.1007/BF00659247. URL: http://link.springer.com/10.1007/BF00659247 (visited on 10/07/2026).

[20] Shu Huang et al. "Multilayer Interface Tracking Model of Pure Tungsten Oxidation". In: *Modelling and Simulation in Materials Science and Engineering* 30.8 (2022). ISSN: 0965-0393 1361-651X. DOI: 10.1088/1361-651X/aca111.

[21] J. Robertson. "The Mechanism of High Temperature Aqueous Corrosion of Stainless Steels". In: *Corrosion Science* 32.4 (Jan. 1991), pp. 443–465. ISSN: 0010938X. DOI: 10.1016/0010-938X(91)90125-9. URL: https://linkinghub.elsevier.com/retrieve/pii/0010938X91901259 (visited on 10/07/2026).

[22] Motoaki Osawa and Fukujirô Ikezu. "Computer Simulation of Intergranular Corrosion of Austenitic Stainless Steel by Means of Finite Element Method". In: *CORROSION ENGINEERING* 34.6 (1985), pp. 331–338. ISSN: 0010-9355. DOI: 10.3323/jcorr1974.34.6_331. URL: http://dx.doi.org/10.3323/jcorr1974.34.6_331.

[23] F.P. Ford. "Quantitative Prediction of Environmentally Assisted Cracking". In: *Corrosion* 52.5 (May 1, 1996), pp. 375–395. ISSN: 0010-9312. DOI: 10.5006/1.3292125. URL: http://dx.doi.org/10.5006/1.3292125.

[24] Tetsuo Shoji, Zhanpeng Lu, and Hiroyoshi Murakami. "Formulating Stress Corrosion Cracking Growth Rates by Combination of Crack Tip Mechanics and Crack Tip Oxidation Kinetics". In: *Corrosion Science* 52.3 (Mar. 2010), pp. 769–779. ISSN: 0010-938X. DOI: 10.1016/j.corsci.2009.10.041. URL: http://dx.doi.org/10.1016/j.corsci.2009.10.041.

[25] C. Z. Fang et al. "A Numerical Study on the Influence of Grain Boundary Oxides on Dwell Fatigue Crack Growth of a Nickel-Based Superalloy". In: *Journal of Materials Science & Technology* 104 (2022), pp. 224–235. ISSN: 10050302. DOI: 10.1016/j.jmst.2021.06.045.